GTPA v2 — Block C, objects and axioms: GTPA-internal questions on the UFD base

Status:

```
document   = GTPA v2 / Block C answers, objects-and-axioms half / READ 2026-09-26 /
             one independent read found one real error (in the crosswalk row
             of EG-02.1, which dropped the conditions this file states; row
             corrected) and seven minor points (applied); not re-read
folder     = gtpa-v2/ (text of the next GTPA edition + its crosswalk from v1.2)
base       = UFD nucleus v0.1, ufd/UFD-NUCLEO.md at research
             7db1e5ef1e1e04daafab7b62538068afafd5dbf5 (blob 4f52a62c):
             convention C1-C6, statements U-01..U-15, Block A answers;
             Block B, gtpa-v2/BLOCK-B-ANSWERS.md at the same commit
             (blob 3d8dea16): statements B-01..B-11
scope      = the objects-and-axioms half of Block C: EG-02.1, EG-02.6,
             EG-10.a3, EG-04.2, EG-05.b, EG-07.b-src, EG-08.a, EG-10.c2;
             and the two HOLDs that follow answered questions: EG-10.a2
             (after EG-02.6) and EG-09.5 (after Block B's EG-09.3).
             The theorems half of Block C (EG-03.1..03.11 and the Thm 9.1
             half of EG-03.0) is a separate file
ceiling    = FORMAL_LOCAL_CONDITIONAL / NONPROBATIVE
claims     = NONE. Claim authority = gtpa-lab/CLAIM-LEDGER.md; nothing here
             is a GTPA claim until it passes the ledger gate (next free ID
             C0138; C0137 is held for EG-05c)
physics    = CLOSED     MATH<->GTPA = UNBOUND     MATH<->UFD = one object only
             (formal-framework/governance/math-owner-binding-record-n0-v1.md),
             not used here
producer   = Claude (Anthropic), workflow writing agent
evaluator  = one independent read, same provider (L-A1), 2026-09-26: FIX_NEEDED
             (EG-02.1 crosswalk row claimed more than §3.1 proves); fixed;
             minor points applied (CO-12 (b) example valid under both
             readings of q_model; CO-05 (a) for any variation of f; F named
             in §3.3; both readings in §3.6; Ch 8 equation typed in §3.7;
             no smoothing claim in §3.9; EG-10.a2 split in §3.0)
checks     = no human check; no cross-provider check; no formal (Lean) check
recompute  = decisive calculations recomputed by machine on 2026-09-26: 31 agree, 3 not mechanizable
```

Opening record. The 29 (+2) questions on GTPA v1.2 were opened by the operator on 2026-09-25 ("3, sí por supuesto"; architecture "Go"), in three blocks. Block A is answered in the UFD nucleus (UFD §3) and Block B in `BLOCK-B-ANSWERS.md`. This file answers the half of Block C that concerns GTPA's own objects and axioms, and the two HOLDs that were waiting on questions now answered. Working style (operator, 2026-09-26): coherent text, one status per statement, one independent read; no tests, check modules or hash pins.

0. How to read this file

- Same format as the UFD nucleus (UFD §0) and Block B. Every statement has an ID `CO-nn`, its types and hypotheses, one status — `DECLARED`, `PROVED_HERE` (complete proof here, checkable by hand), `STANDARD_FACT` (exact locator) or `OPEN` — and one line in plain Spanish (`En llano`).
- Section 3 answers ten questions: eight questions and two HOLDs. Each has the exact question text of the triage record (research `d0cf83e`), the GTPA v1.2 locators, exactly one verdict — `HOLDS_UNDER_STATED_HYPOTHESES` / `FAILS_WITH_COUNTEREXAMPLE` / `ILL_POSED_AS_WRITTEN` (+ the repair GTPA v2 adopts) / `OPEN` — the argument, the repair, and one sentence in plain Spanish. The lead adds one row per answer to `CROSSWALK-v1.2.md`; this file does not edit it.

- The base motion is UFD's: the first-order Itô SDE (M) on R^n and, under the pairing (P), its paired variant (M_P) (C1–C5; U-01, U-02, U-08). The notation of UFD §0 and of Block B §1 is used unchanged; §1 adds what this half needs.
- Words: UFD's rule applies. Words from physics appear only inside quotations or paraphrases of v1.2 and in explicit negations. v1.2's names (M_{cog}, p_{est}, P_{cat}, S_{total}, `Work_Term`, "Regime B", "Beta-Switch") are used as names of v1.2 objects and nothing more.
- Page numbers are PDF page indices of GTPA v1.2 (`quasar:modulos/formacion/fuentes/GTPA-v1.2.pdf`, SHA-256 $0c9434c1...865f$), the numbering of its text export (blob 32787369) and of the triage. Quotations were checked against the text export; the lines whose signs decide a verdict (pp.8, 44, 45, 46, 78, 79) were also checked on the rendered PDF.
- No MATH object is used as proof anywhere in this file. No TECD text was needed.

1. Notation added for this half

- $\beta > 0$ is v1.2's weight of $L_{\text{informational}}$ in S_{total} (Ax 4.2, p.30). In GTPA v2 it is a declared positive constant, with no reading as "k_B * T_substrate * ln(2)" (Block B, EG-02.5 repair: GTPA v2 has no temperature).
- Riemannian notation, used only for App E (CO-02 to CO-06). M is a smooth manifold of dimension $d \geq 2$, closed (compact, without boundary) unless said otherwise; g is a smooth Riemannian metric on it (v1.2's G_{model}, G_{ij}; UFD's letter G is the mobility and is not used there); ∇ is its Levi-Civita connection; $\mathrm{Ric} = \mathrm{Ric}(g)$, with components R_{ij}; R is the scalar curvature; K the Gaussian curvature when $d = 2$; $\mathrm{Hess}\, f \coloneqq \nabla\nabla f$ (v1.2's $\mathrm{Hessian}_{ij}(f)$); $\Delta f \coloneqq \mathrm{tr}_g \mathrm{Hess}\, f$; dV is the volume form. For symmetric 2-tensors v, a, b: $\mathrm{tr}\, v \coloneqq g^{ij} v_{ij}$, $\langle a, b\rangle \coloneqq g^{ik} g^{jl} a_{ij} b_{kl}$, $|a|^2 \coloneqq \langle a, a\rangle$. The clock of a metric flow is s.
- For a real x: $x^+ \coloneqq \max(x, 0)$, $x^- \coloneqq \max(-x, 0)$, so $x = x^+ - x^-$.
- v1.2's discrepancy components `epsilon_q`, `epsilon_p` (Axiom 5.1, p.33) are written e_q, e_p, as in v1.2's own summary (p.7); ε stays UFD's noise level (C4). v1.2's filter threshold "epsilon" (Thm 7.1, p.44) is written θ.
- Sums are written `sum`, as in UFD §0 (Σ is the noise matrix).

2. Statements

CO-01 — The minimizer of Action_Local and the three printed forms of u_opt `[PROVED_HERE]` (EG-02.1)

Setting. Integers $k, N \geq 1$; $B \in R^{N\times k}$; $G_{\text{reg}} \in R^{N\times N}$ symmetric with $G_{\text{reg}} \succeq 0$ (as its use as a norm requires, B-08 (d)); $\lambda \coloneqq \lambda_{\text{effort}} > 0$; $F \coloneqq F_{\text{req}} \in R^N$. The only typing used is that F_{req} and Bu lie in one R^N on which G_{reg} is a symmetric form. (v1.2 lets B map into "the Phase Space Tangent Bundle T(T*Q)", p.42, while F_{req} is a gradient in p, p.43; this file does not settle that.) Put

$$AL(u) \coloneqq \tfrac{1}{2}(F - Bu)^{\mathsf{T}} G_{\text{reg}} (F - Bu) + \tfrac{1}{2}\lambda |u|^2 \qquad (v1.2's\ \mathrm{Action}_{\text{Local}},\ p.43),$$
$$Z \coloneqq B^{\mathsf{T}} G_{\text{reg}} B + \lambda I \qquad (\text{Def } 7.2,\ p.44).$$

(a) $\nabla AL(u) = Zu - B^{\mathsf{T}} G_{\text{reg}} F$ and $D^2 AL = Z$. Moreover $Z \succeq \lambda I \succ 0$, so AL is strictly convex and has exactly one minimizer, $u^* = Z^{-1} B^{\mathsf{T}} G_{\text{reg}} F$.

(b) For every $u_0 \in R^k$, $u^* = u_0 - Z^{-1} \nabla AL(u_0)$: one Newton step from any point reaches the minimizer. In particular $u^* = Z^{-1}(-\nabla AL(0))$.

(c) $Z^{-1}\,\nabla AL(u_0) = u_0 - u^*$. It equals u^* iff $u_0 = 2u^*$, and at $u_0 = 0$ it equals $-u^*$. For every evaluation point u_0 fixed in advance (not depending on F), some data make it differ from the minimizer: with $k = N = 1$ and $B = G_{\text{reg}} = \lambda = 1$, $AL(u) = ½(F-u)^2 + ½u^2$, $Z = 2$, $u^* = F/2$ and $Z^{-1}\,AL'(u_0) = u_0 - F/2$, which differs from $F/2$ whenever $F \neq u_0$. For $u_0 = 0$ and $F = 1$ it is $-½$, where $AL(-½) = 5/4$, while the minimum is $AL(½) = 1/4$.

(d) If $Z = U\,S\,V^{\mathsf{T}}$ is a singular value decomposition, then $V\,S^{-1}\,U^{\mathsf{T}} = Z^{-1}$, so $V\,S^{-1}\,U^{\mathsf{T}}\,B^{\mathsf{T}}\,G_{\text{reg}}\,F = u^*$. Every singular value of Z is $\geq \lambda$; a filter that inverts only the singular values $> \theta$ inverts all of them when $\theta < \lambda$.

Proof. (a) Since G_{reg} is symmetric, $AL(u) = ½F^{\mathsf{T}}G_{\text{reg}}F - u^{\mathsf{T}}B^{\mathsf{T}}G_{\text{reg}}F + ½u^{\mathsf{T}}(B^{\mathsf{T}}G_{\text{reg}}B + \lambda I)u$; differentiate. $u^{\mathsf{T}}B^{\mathsf{T}}G_{\text{reg}}Bu = (Bu)^{\mathsf{T}}G_{\text{reg}}(Bu) \geq 0$, so $u^{\mathsf{T}}Zu \geq \lambda|u|^2$. A C^2 function whose Hessian is everywhere positive definite is strictly convex; its gradient vanishes exactly at u^*, which is therefore its unique minimizer. (b) $Zu^* = B^{\mathsf{T}}G_{\text{reg}}F$, so $\nabla AL(u_0) = Z(u_0 - u^*)$. (c) From (b); the numbers are direct. (d) U, V are orthogonal and S is diagonal and invertible, so $(USV^{\mathsf{T}})^{-1} = VS^{-1}U^{\mathsf{T}}$. $Z = Q\Lambda Q^{\mathsf{T}}$ with Q orthogonal and $\Lambda \geq \lambda I$ diagonal is a singular value decomposition, and the singular values of a matrix do not depend on the decomposition. ∎

En llano: el coste local de v1.2 tiene un único mínimo, $u^* = Z^{-1} \cdot B^{\mathsf{T}} \cdot G_{\text{reg}} \cdot F_{\text{req}}$, al que se llega con un solo paso de Newton desde cualquier punto (restando Z^{-1} por el gradiente); la fórmula del Teorema 7.1, sin el signo menos, da justo $-u^*$ si el gradiente se toma en cero, y la del paso 3 (con valores singulares) da u^* cuando el filtro no quita nada.

CO-02 — Ricci flow: the object, the facts used, and where it is well posed `[DECLARED + STANDARD_FACT]` (EG-02.6, EG-10.a2)

(a) `DECLARED`. A Ricci flow on M over a clock interval $[0, T)$ is a smooth family $g(s)$ of Riemannian metrics with

$$\partial_s\, g = -2\ \mathrm{Ric}(g(s))$$

(Hamilton's flow; the equation v1.2 prints as "d(G_ij)/dt = -2 * R_ij", p.79).

(b) `STANDARD_FACT` [do Carmo 1992, Ch. 2 (Levi-Civita connection), Ch. 4 (Ricci and scalar curvature), Ch. 8 (constant curvature)]. The facts used below: (i) for a constant $c > 0$, cg has the same Levi-Civita connection as g (the Christoffel symbols do not change when g is multiplied by a constant), hence $\mathrm{Ric}(cg) = \mathrm{Ric}(g)$, $\mathrm{Hess}_{cg}\, f = \mathrm{Hess}_g\, f$, and $K(cg) = K(g)/c$ when $d = 2$; (ii) for a diffeomorphism φ, $\mathrm{Ric}(\varphi^* g) = \varphi^*\,\mathrm{Ric}(g)$; (iii) when $d = 2$, $\mathrm{Ric} = K\,g$, and the unit round sphere $S^2 \subset R^3$ has $K = 1$, so $\mathrm{Ric} = g$; (iv) on the unit sphere the restriction z of the third coordinate satisfies $\mathrm{Hess}\, z = -z\,g$. [(iv) directly: $\nabla z(x) = e_3 - z(x)\,x$, the tangential part of e_3; for a tangent vector w at x the derivative of this field along w in R^3 is $-dz(w)\,x - z(x)\,w$, whose tangential part is $-z(x)\,w$.]

(c) `STANDARD_FACT`. If M is closed and g_0 is smooth, there are $T > 0$ and exactly one Ricci flow on $[0, T)$ with $g(0) = g_0$ [Hamilton 1982; DeTurck 1983]. If M is not compact (for instance R^d) and g_0 is complete with bounded curvature, a Ricci flow with bounded curvature exists on some $[0, T)$ [Shi 1989], and it is unique among complete solutions with bounded curvature [Chen–Zhu 2006]. Outside these classes this file claims nothing; in particular nothing is claimed on v1.2's own setting, where $M_{\text{model}} \subset T^*Q$ (Ax 2.1, p.21) and Q is "a Sobolev space" (Ax 1.3, p.18).

En llano: el "flujo de Ricci" es una regla matemática conocida que deforma poco a poco una métrica según su curvatura; en una forma cerrada y lisa, o en el espacio con una métrica completa de curvatura acotada, existe y es única durante un tiempo; sobre el marco de v1.2 (un espacio de Sobolev) no se afirma nada.

CO-03 — The rate is a clock, and a term proportional to the metric is a rescaling `[PROVED_HERE]` (EG-02.6, EG-10.a3)

(a) Let $c > 0$ be a constant. A family $g(s)$ solves $\partial_s g = -2\,\mathrm{Ric}(g)$ iff $h(t) := g(ct)$ solves $\partial_t h = -2c\,\mathrm{Ric}(h)$. With $c = 1/(2\beta)$ the second equation is E.4's "dG/dt ~ - (1 / beta) * R_ij" (p.79) read as an equality. So the coefficients -2 (Ax 6.2, App E) and $-1/\beta$ (E.4) give the same metrics on two clocks, $t = 2\beta s$.

(b) Let $f(s)$ be smooth functions on M and ρ a continuous function of time, and put $a(t) := \exp(\int_0^t \rho)$ and $\sigma(t) := \int_0^t a^{-1}$. If $g(s)$ solves $\partial_s g = -2(\mathrm{Ric}(g) + \mathrm{Hess}_g f(s))$, then $\tilde{g}(t) := a(t)\, g(\sigma(t))$ solves

$$\partial_t \tilde{g} = -2\,(\mathrm{Ric}(\tilde{g}) + \mathrm{Hess}_{\tilde{g}} f(\sigma(t))) + \rho(t)\,\tilde{g},$$

and every solution $\tilde{g}$ of this equation arises in this way, from $g(\sigma(t)) := \tilde{g}(t)/a(t)$. So adding or dropping a term proportional to the metric ("the trace rescaling term", p.78) changes a solution only by a time-dependent constant factor and a change of clock.

Proof. By CO-02 (b)(i), Ric and Hess f do not change when the metric is multiplied by a positive constant, and at each fixed t, $\tilde{g}(t)$ is such a multiple of $g(\sigma(t))$. (a) Chain rule. (b) $\partial_t \tilde{g} = a'\, g(\sigma) + a\,\sigma'\,\partial_s g(\sigma) = \rho\,\tilde{g} - 2(\mathrm{Ric}(\tilde{g}) + \mathrm{Hess}_{\tilde{g}} f(\sigma))$, because $a'/a = \rho$ and $a\sigma' = 1$. Conversely, with $g(\sigma(t)) := \tilde{g}(t)/a(t)$: $\sigma'\,\partial_s g = \partial_t(\tilde{g}/a) = (\partial_t \tilde{g} - \rho\,\tilde{g})/a = -2(\mathrm{Ric}(\tilde{g}) + \mathrm{Hess}_{\tilde{g}} f)/a$; since $\sigma' = 1/a$ and $g = \tilde{g}/a$ at each time, $\partial_s g = -2(\mathrm{Ric}(g) + \mathrm{Hess}_g f)$. ∎

En llano: cambiar el -2 por $-1/\beta$ solo cambia la unidad del reloj; y quitar un término proporcional a la propia métrica solo reescala la métrica por un factor en cada instante y cambia el reloj, así que "ignorar" ese término es legítimo salvo ese reescalado.

CO-04 — Two readings of Ax 6.2 are two different flows `[PROVED_HERE]` (EG-02.6)

Let E be a fixed Riemannian metric on M. Reading (R_E): `Error_Tensor` is E, and the flow is $\partial_s g = -2\,\mathrm{Ric}(E)$. Reading (R_g): `Error_Tensor` is the model metric g itself (or a constant multiple of it, CO-02 (b)(i)), and the flow is the Ricci flow $\partial_s g = -2\,\mathrm{Ric}(g)$.

(a) Under (R_E) the right side does not depend on g: the only solution is $g(s) = g(0) - 2s\,\mathrm{Ric}(E)$, for as long as it stays positive definite.

(b) Example. $M = S^2$, E the unit round metric ($\mathrm{Ric}(E) = E$, CO-02 (b)(iii)), φ a diffeomorphism of S^2 that is not an isometry of E (for instance the map $z \mapsto 2z$ in a stereographic coordinate), and $g(0) := \varphi^* E$. Under (R_g) the flow is $g(s) = (1 - 2s)\,\varphi^* E$ for $0 \le s < ½$; under (R_E) it is $g(s) = \varphi^* E - 2sE$. They differ at every $s > 0$ at which both are defined, because $\varphi^* E \neq E$.

Proof. (a) Integrate in s. (b) $\mathrm{Ric}((1 - 2s)\varphi^* E) = \mathrm{Ric}(\varphi^* E) = \varphi^*\,\mathrm{Ric}(E) = \varphi^* E$ (CO-02 (b)(i), (ii)) and $\partial_s[(1 - 2s)\varphi^* E] = -2\varphi^* E$; by the uniqueness in CO-02 (c) this is the Ricci flow from $\varphi^* E$. The difference of the two solutions is $2s(E - \varphi^* E)$. The map $z \mapsto 2z$ is not an isometry: in the stereographic coordinate $E = 4|dz|^2/(1 + |z|^2)^2$, and at $z = 0$ the pulled-back metric is $16|dz|^2$, four times E. ∎

En llano: si el "tensor de error" es la propia métrica del modelo, la regla del capítulo 6 es el flujo de Ricci; si es otra métrica fija, es una regla distinta, y en un ejemplo sencillo sobre la esfera las dos dan métricas distintas desde el primer instante.

CO-05 — The first variation of Perelman's functionals, and App E's Steps 2 and 3 `[PROVED_HERE + STANDARD_FACT]` (EG-10.a3, EG-02.6, EG-10.a2)

Setting. M closed, g a metric, $f \in C^\infty(M)$, $d\nu := e^{-f}\,dV$, $\tau > 0$ a constant. Perelman's functionals are

$$F_P(g, f) \quad := \int_M (R + |\nabla f|^2)\,d\nu,$$

$$W_P(g, f, \tau) := \int_M [\tau\,(R + |\nabla f|^2) + f - d]\, dv.$$

W_P is v1.2's W (Axiom E.1, p.77) with $u^2 = e^{-f}$; Perelman's factor $(4\pi\tau)^{-d/2}$ is dropped, which at fixed τ only multiplies W_P and its variations by a positive constant. A variation is a smooth family (g_r, f_r) with $\partial_r g|_{r=0} = v$ and $\partial_r f|_{r=0} = h$; it keeps dv fixed (to first order) iff $h = \frac{1}{2}\operatorname{tr} v$.

(a) PROVED_HERE. For every variation that keeps dv fixed,

$$\delta F_P = -\int_M \langle v, \operatorname{Ric} + \operatorname{Hess} f\rangle\, dv, \qquad \delta W_P = -\tau \int_M \langle v, \operatorname{Ric} + \operatorname{Hess} f - g/(2\tau)\rangle\, dv.$$

These are Perelman's formulas [Perelman 2002, §1.1 and §3.1, with the constraint $h = \frac{1}{2}\operatorname{tr} v$]. In the gradient of W_P for the pairing $\int\langle\cdot,\cdot\rangle\, dv$, namely $-\tau(\operatorname{Ric} + \operatorname{Hess} f - g/(2\tau))$, the terms R_{ij} and $\nabla_i\nabla_j f$ carry the same sign. For any other choice of the variation h of f (a function of v, or independent of it), the extra terms of δF_P and δW_P are of the form $\int \varphi \operatorname{tr} v\, dv = \int \langle v, \varphi g\rangle\, dv$ or $\int \psi h\, dv$, with functions φ, ψ: they add to the part paired with v only a function times g, so the relative sign of Ric and Hess f does not change.

(b) PROVED_HERE. App E's Step 3 tensor $-2(\operatorname{Ric} + \operatorname{Hess} f)$ is not a real multiple of Step 2's printed tensor $-\operatorname{Ric} + \operatorname{Hess} f - a\,g$, neither with the printed $a = 1/(2\tau)$ nor with $a = 0$ (the metric term dropped). Example: $M = S^2$ with the unit round metric, $f = z^2$, at the point $x = (1, 0, 0)$. In the orthonormal frame (e_2, e_3) of $T_x S^2$: $g = I$, $\operatorname{Ric} = I$, $\operatorname{Hess} f = \operatorname{diag}(0, 2)$; Step 3's tensor is $\operatorname{diag}(-2, -6)$ and Step 2's is $\operatorname{diag}(-1 - a, 1 - a)$; a real multiple would need $a = -2$.

(c) PROVED_HERE. Along any smooth family $(g(s), f(s))$ on M with $\partial_s g = -2(\operatorname{Ric} + \operatorname{Hess} f)$ (Step 3) and $\partial_s f = -R - \Delta f$ (the choice that keeps dv fixed),

$$d/ds\, F_P = 2 \int_M |\operatorname{Ric} + \operatorname{Hess} f|^2\, dv \quad \geq 0.$$

So Step 3's flow is twice the gradient of F_P for the pairing $\int\langle\cdot,\cdot\rangle\, dv$: an ascent direction, not a descent. [For W_P, the analogous monotonicity, with τ decreasing at unit rate, is Perelman 2002, §3.1.]

(d) STANDARD_FACT. $\operatorname{Hess} f = \frac{1}{2} L_{\nabla f}\, g$ (L the Lie derivative), so Step 3's flow is $\partial_s g = -2\ \operatorname{Ric} - L_{\nabla f}\, g$; pulled back by the diffeomorphisms that ∇f generates, its solutions become Ricci flows [Perelman 2002, §1]. This is Thm E.1's "Lie Derivative" reading (p.79), which is right.

Proof. (a) The two identities of App E's Step 1 (p.78) are correct and standard (Perelman uses them in §1.1): $\delta(dV) = \frac{1}{2}(\operatorname{tr} v)\, dV$ and $\delta R = -\Delta(\operatorname{tr} v) + \nabla^i\nabla^j v_{ij} - \langle \operatorname{Ric}, v\rangle$. Also $\delta|\nabla f|^2 = -v^{ij}\,\partial_i f\,\partial_j f + 2\langle \nabla f, \nabla h\rangle$ and $\delta(dv) = (\frac{1}{2}\operatorname{tr} v - h)\, dv$, which is 0 under the constraint. Hence

$$\delta F_P = \int [-\Delta(\operatorname{tr} v) + \nabla^i\nabla^j v_{ij} - \langle \operatorname{Ric}, v\rangle - v^{ij}\,\partial_i f\,\partial_j f + 2\langle \nabla f, \nabla h\rangle]\, dv.$$

Integrate by parts on the closed M, using $\Delta(e^{-f}) = (|\nabla f|^2 - \Delta f)\, e^{-f}$, $\nabla^i\nabla^j(e^{-f}) = (\nabla^i f\,\nabla^j f - \nabla^i\nabla^j f)\, e^{-f}$ and $\operatorname{div}(e^{-f}\nabla f) = (\Delta f - |\nabla f|^2)\, e^{-f}$: $\int -\Delta(\operatorname{tr} v)\, dv = -\int (\operatorname{tr} v)(|\nabla f|^2 - \Delta f)\, dv$; $\int \nabla^i\nabla^j v_{ij}\, dv = \int (v_{ij}\,\nabla^i f\,\nabla^j f - \langle v, \operatorname{Hess} f\rangle)\, dv$; $\int 2\langle \nabla f, \nabla h\rangle\, dv = 2\int h\,(|\nabla f|^2 - \Delta f)\, dv$. The terms $v_{ij}\,\nabla^i f\,\nabla^j f$ cancel, and what remains is $\delta F_P = -\int \langle v, \operatorname{Ric} + \operatorname{Hess} f\rangle\, dv + \int (2h - \operatorname{tr} v)(|\nabla f|^2 - \Delta f)\, dv$, whose last integral is 0. For W_P at fixed τ: $\delta \int (f - d)\, dv = \int h\, dv = \int \langle v, g/2\rangle\, dv$, so $\delta W_P = \tau\,\delta F_P + \int \langle v, g/2\rangle\, dv$, which is the displayed formula. Without the constraint, the same computation (the integrations by parts hold for any h) adds, to δF_P, $\int (2h - \operatorname{tr} v)(|\nabla f|^2 - \Delta f)\, dv$ and $\int (R + |\nabla f|^2)(\frac{1}{2}\operatorname{tr} v - h)\, dv$; to δW_P, τ times these, $\int (f - d)(\frac{1}{2}\operatorname{tr} v - h)\, dv$ and $\int h\, dv - \int \langle v, g/2\rangle\, dv$. Since $\langle v, g\rangle = \operatorname{tr} v$, each is of the form $\int \varphi \operatorname{tr} v\, dv$ or $\int \psi h\, dv$. (b) By CO-02 (b)(iii)–(iv), $\operatorname{Ric} = g$ and $\operatorname{Hess}(z^2) = 2\, dz \otimes dz + 2z \operatorname{Hess} z = 2\, dz \otimes dz - 2z^2 g$, which at x (where $z = 0$ and $\nabla z = e_3$) is $\operatorname{diag}(0, 2)$. Then $-2(\operatorname{Ric} + \operatorname{Hess} f) = \operatorname{diag}(-2, -6)$ and $-\operatorname{Ric} + \operatorname{Hess} f - a\,g = \operatorname{diag}(-1 - a, 1 - a)$. If $\operatorname{diag}(-2, -6) = c \cdot \operatorname{diag}(-1 - a, 1 - a)$, then $c \neq 0$, and the ratio of the two entries gives $3 = (1 - a)/(-1 - a)$, i.e. $a = -2$. (c) Apply (a) with $v = -2(\operatorname{Ric} + \operatorname{Hess} f)$ and $h = -R - \Delta f$: then

½ tr $v = -R - \Delta f = h$, so dv is fixed, and $\delta F_P = 2 \int |\mathrm{Ric} + \mathrm{Hess}\, f|^2\, dv$; with v free, (a) says that the gradient of F_P is $-(\mathrm{Ric} + \mathrm{Hess}\, f)$. (d) $(L_X\, g)_{ij} = \nabla_i X_j + \nabla_j X_i$; with $X = \nabla f$ this is $2\,\nabla_i \nabla_j f$. ∎

En llano: hecha bien, la cuenta del apéndice E da la curvatura de Ricci y el término de f con el mismo signo; el paso 2 impreso los trae con signos opuestos, así que el paso 3 (que es la ecuación conocida) no sale de él; y esa ecuación hace subir, no bajar, la cantidad F de Perelman.

CO-06 — The flow can end in finite time `[PROVED_HERE]` (EG-10.a2)

$M = S^2$ and g_0 the unit round metric. The Ricci flow from g_0 is $g(s) = (1 - 2s)\, g_0$ for $0 \le s < ½$. Its Gaussian curvature is $K(s) = 1/(1 - 2s)$ and its area is $4\pi(1 - 2s)$: as $s \to ½$ the curvature tends to $+\infty$ and the area to 0.

Proof. $\mathrm{Ric}((1 - 2s)g_0) = \mathrm{Ric}(g_0) = g_0 = -½\,\partial_s[(1 - 2s)g_0]$ (CO-02 (b)(i), (iii)); uniqueness is CO-02 (c); $K(c\, g_0) = 1/c$ and the area is multiplied by c. ∎

So "Ricci Flow diffuses curvature" and "the flow naturally spreads this curvature out" (E.4, p.79), and "smooth out singularities" (Ax 6.2, p.41), are not unconditional: from a smooth metric of constant curvature the flow ends in finite time, with curvature tending to infinity.

En llano: el flujo de Ricci no siempre "suaviza": a la esfera redonda la encoge hasta un punto en un tiempo finito, y la curvatura se dispara.

CO-07 — Mode frequencies need a declared pair of matrices `[PROVED_HERE]` (EG-04.2)

(a) Let M, H be symmetric $d \times d$ matrices with $M \succ 0$ and $H \succ 0$. $M^{-1}H$ is similar to $M^{-1/2}\, H\, M^{-1/2} \succ 0$, so it has d positive eigenvalues $\lambda_1, \ldots, \lambda_d$ (with multiplicity). Define the multiset

$$\mathrm{Spec}(M, H) \coloneqq \{\lambda_1^{1/2}, \ldots, \lambda_d^{1/2}\}\ :$$

the frequencies of the modes of $M\,\ddot{x} + H\,x = 0$ (B-08 (b), with no first-derivative term). For every invertible L, $\mathrm{Spec}(L^{\mathsf{T}}ML, L^{\mathsf{T}}HL) = \mathrm{Spec}(M, H)$. Spec depends on the pair: for $d = 1$, $\mathrm{Spec}(m, h) = \{(h/m)^{1/2}\}$, so with either entry fixed it takes every positive value.

(b) On the model side v1.2 does not fix the second matrix. Take Ax 6.1 (p.37) with constant M_{cog} (Block B, EG-02.4 repair: a declared symmetric positive definite matrix) and constant G_{model}, `Gamma_cog = 0`, the base state held fixed, and `F_cognitive = - grad( beta * E_Hamiltonian )` (p.37), with $E_{\mathrm{Hamiltonian}} = ½(\langle e_q, G_{\mathrm{model}}\, e_q\rangle + \langle e_p, M_{\mathrm{cog}}^{-1}\, e_p\rangle)$ (Def 5.1, p.34), $e_q = \Pi(q) - q_{\mathrm{est}}$, $e_p = J_\Pi(q)\, p - p_{\mathrm{est}}$, and "grad" taken in q_{est}. Then Ax 6.1 reads $M_{\mathrm{cog}}\, \ddot{q}_{\mathrm{est}} = -H\,(q_{\mathrm{est}} - \Pi(q)) + \kappa$, with κ the (here constant) `Correction_Ito`, and

- $H = \beta\, G_{\mathrm{model}}$ if "grad" is the differential in q_{est} (the reading in which $F_{\mathrm{cognitive}}$ has the type of dp_{est}/dt, hence of $p_{\mathrm{est}} = M_{\mathrm{cog}}\, \dot{q}_{\mathrm{est}}$, as `dq_est/dt = M_cog^(-1) * p_est` requires);
- $H = \beta\, I$ if "grad" is the gradient for the metric G_{model}.

The mode frequencies are $\mathrm{Spec}(M_{\mathrm{cog}}, \beta G_{\mathrm{model}})$ and $\mathrm{Spec}(M_{\mathrm{cog}}, \beta I)$ respectively; for $d = 1$, $M_{\mathrm{cog}} = m$ and $G_{\mathrm{model}} = \gamma \neq 1$ they are $(\beta\gamma/m)^{1/2} \neq (\beta/m)^{1/2}$.

(c) On the base side (UFD, first order, C5) there is no second-order structure. At a critical point q^* of J with $H^* \coloneqq D^2 J(q^*) \succ 0$, the noiseless linearization is $\dot{x} = (A^* - G^*)\, H^*\, x$ (B-08 (c)). If $A^* = 0$, all its eigenvalues are real and ≤ 0: there is no oscillation at all. With a circulating part there can be nonreal eigenvalues; in B-08 (c)'s example they are $-k(g \pm ia)$, with rotation rate $k|a|$.

(d) Multisets of different sizes are never equal. For $d \neq n$ the candidate readings of "equal frequencies" give different verdicts on the same pair: for a model spectrum $\{1\}$ and a base spectrum $\{1, 2\}$, "equal as

sets" is false, "model contained in base" is true, "equal lowest frequencies" is true, and "equal highest frequencies" is false.

Proof. (a) $M^{-1}H = M^{-1/2}(M^{-1/2}HM^{-1/2})M^{1/2}$ and $(L^{\mathsf{T}}ML)^{-1}(L^{\mathsf{T}}HL) = L^{-1}(M^{-1}H)L$. (b) With M_{cog} constant, $p_{\text{est}} = M_{\text{cog}}\,\dot{q}_{\text{est}}$, so Ax 6.1 gives `M_cog q̈_est = F_cognitive + Correction_Ito` (`Gamma_cog = 0`). The part of $E_{\text{Hamiltonian}}$ with e_p does not depend on q_{est}. For the part with e_q, with G_{model} constant, the differential in q_{est} is $-G_{\text{model}}\,e_q = G_{\text{model}}\,(q_{\text{est}} - \Pi(q))$, and the gradient for the metric G_{model} is G_{model}^{-1} times it, $q_{\text{est}} - \Pi(q)$. `Correction_Ito` depends on the base state only (B-01), so it is constant. (c) ${H^*}^{1/2}(G^*H^*){H^*}^{-1/2} = {H^*}^{1/2}\,G^*\,{H^*}^{1/2}$ is symmetric and $\succeq 0$, so G^*H^* has real eigenvalues ≥ 0. (d) By inspection. ∎

En llano: una forma sola no tiene frecuencias: hacen falta dos matrices (la que multiplica a la segunda derivada y la de rigidez) en un punto de equilibrio; v1.2 no fija la de rigidez del modelo (depende de cómo se lea su "gradiente"), la base de primer orden de UFD no oscila salvo que haya giro, y con tamaños distintos "igualdad de frecuencias" admite lecturas que dan respuestas distintas.

CO-08 — A binary choice that minimizes an integral cost `[PROVED_HERE]` (EG-05.b)

Setting. $T > 0$; $\beta > 0$; $c_{\text{act}}, c_{\text{inf}} : [0,T] \to [0,\infty)$ integrable. A mode path is a measurable $m : [0, T] \to \{\text{act}, \text{inf}\}$, with cost

$$\text{Cost}(m) := \int_0^T \ell(t, m(t))\,dt, \qquad \ell(t,\text{act}) := c_{\text{act}}(t), \qquad \ell(t,\text{inf}) := \beta\,c_{\text{inf}}(t).$$

(In Ax 7.2's terms: c_{act} is $||\,u_{\text{opt}}\,||^2_{\text{Metric}}$; c_{inf} is $||\,dp_{\text{est}}/dt\,||^2$ without β; β is Ax 4.2's weight.)

(a) Hypothesis (X): c_{act} and c_{inf} do not depend on m. Under (X), the path $m^*(t) :=$ act if $c_{\text{act}}(t) < \beta\,c_{\text{inf}}(t)$, and inf otherwise, minimizes `Cost`, and every minimizer equals m^* at almost every t with $c_{\text{act}}(t) \neq \beta\,c_{\text{inf}}(t)$.

(b) Under (X), β enters once, on the inference side. v1.2's p.45 rule (β inside `Unified_Cost_Inference`, then "Cost_Control < Cost_Inference") is the rule of (a). p.8's rule "Cost_Control < beta * Cost_Inference" is the rule of (a) if its `Cost_Inference` is the cost without β; if it is p.45's `Unified_Cost_Inference`, p.8 compares $c_{\text{act}} < \beta^2\,c_{\text{inf}}$, a different rule: with $\beta = 2$, $c_{\text{inf}} = 1$ and $c_{\text{act}} = 3$, p.45 gives inf ($3 < 2$ is false), and p.8 read this way gives `act` ($3 < 4$).

(c) Without (X) the rule of (a) need not minimize. Two steps of unit length: at step 1, $c_{\text{act}} = 1$ and $\beta\,c_{\text{inf}} = 2$; at step 2 both modes cost 10 if step 1 was `act`, and 0 if step 1 was inf. The rule chooses `act` at step 1 and pays 11; choosing inf at step 1 pays 2.

Proof. (a) For every t, $\ell(t, m(t)) \geq \min(c_{\text{act}}(t), \beta c_{\text{inf}}(t)) = \ell(t, m^*(t))$; integrate. m^* is measurable because $\{c_{\text{act}} < \beta c_{\text{inf}}\}$ is. If $\text{Cost}(m) = \text{Cost}(m^*)$, the nonnegative function $\ell(\cdot, m(\cdot)) - \ell(\cdot, m^*(\cdot))$ has integral 0, so it vanishes almost everywhere; where $c_{\text{act}} \neq \beta c_{\text{inf}}$ this gives $m = m^*$. (b), (c) By inspection. ∎

En llano: si los costes de actuar y de pensar no dependen de lo que se eligió antes, elegir en cada instante lo más barato (con β una sola vez, del lado de pensar) da el coste total mínimo; si lo que se elige ahora cambia los costes de después, esa regla puede fallar.

CO-09 — The regeneration term of §7.4 `[PROVED_HERE]` (EG-07.b-src)

For a real P (v1.2's `<F, v>`) and $\eta \in [0,1]$ (v1.2's `eta_regen`), v1.2's term is `Work_Term(P, η) := P` if $P > 0$, and $(1-\eta)\,P$ otherwise (p.46).

(a) `Work_Term(P, η) = P⁺ − (1 − η) P⁻`. It is a function of (P, η) alone: no constraint potential enters it, so it is the same function with the soft potential of Regime A (p.34) and with the logarithmic barrier of Regime B (p.35).

(b) Read as a running cost, it credits a phase with $P < 0$ (v1.2's "braking phase") by $(1-\eta)|P|$: the fraction $1-\eta$ of $|P|$, all of it when $\eta = 0$ and nothing when $\eta = 1$. The sentence it is meant to formalize, "recover a fraction eta_regen" (p.46), is the cost $P^+ - \eta P^-$. Read instead as a loss, it should be ≥ 0, but `Work_Term(P, η) < 0` when $P < 0$ and $\eta < 1$.

(c) These four readings of v1.2 are jointly inconsistent: (i) regeneration is "Regime B only" (p.46); (ii) a biological agent has it ("ATP recovery in biology", p.46); (iii) Regime B agents are Class B ("Regime B: The Logarithmic Barrier (Class B)", p.35); (iv) no biological agent is Class B (Class B's constraints are "physically absent in biology", p.56; "Class A agents model humans and animals", p.75). From (ii), (i) and (iii), some biological agent is Class B, against (iv).

Proof. (a) Check the cases $P > 0$ and $P \leq 0$; the formula contains only P and η. (b) From (a). (c) Propositional logic. ∎

En llano: el término de "regeneración" no usa para nada la barrera, así que es el mismo en los dos regímenes; tal como está escrito, leído como coste recupera la fracción contraria a la que dice el texto, y leído como pérdida tiene el signo cambiado; y el ejemplo biológico choca con lo que el propio texto dice de la biología.

CO-10 — Joining paired agents gives a paired motion `[PROVED_HERE]` (EG-08.a)

For $i = 1..N$ let (A_i, G_i, Σ_i) be coefficients on R^{n_i}, functions of q_i, that satisfy the pairing (P) of U-08 with the same $\varepsilon > 0$, and let $W_1, \ldots, W_N$ be independent standard Brownian motions of dimensions m_i. Put $n := \sum_i n_i$, $q := (q_1, \ldots, q_N) \in R^n$, $A := \mathrm{diag}(A_i)$, $G := \mathrm{diag}(G_i)$, $\Sigma := \mathrm{diag}(\Sigma_i)$, $W := (W_1, \ldots, W_N)$, and let $J \in C^2(R^n)$ be any joint landscape, for instance $J(q) = \sum_i J_i(q_i) + J_{\mathrm{int}}(q)$. Then $(A, G, \Sigma, \varepsilon)$ satisfies (P) on R^n: the joint equation

$$dq = [(A - G)\nabla J + \varepsilon\ \mathrm{div}\, G](q)\, dt + \Sigma(q)\, dW$$

is a paired motion (M_P) on R^n (under (H)), and $e^{-J/\varepsilon}$ is infinitesimally invariant for it (U-08 (a)). With $J_{\mathrm{int}} = 0$, its i-th block is agent i's own paired motion.

Proof. A is antisymmetric; G is symmetric and $\succeq 0$. The column divergence of A and the divergence of G are computed block by block, so $\mathrm{div}_c A = (\mathrm{div}_c A_i)_i = 0$ and $\mathrm{div}\, G = (\mathrm{div}\, G_i)_i$. $\Sigma\Sigma^{\mathsf{T}} = \mathrm{diag}(\Sigma_i \Sigma_i^{\mathsf{T}}) = \mathrm{diag}(2\varepsilon G_i) = 2\varepsilon G$, and W is a standard Brownian motion. With $J_{\mathrm{int}} = 0$, $\nabla J = (\nabla J_i)_i$, so the drift splits block by block. ∎

En llano: juntar varios agentes emparejados, con el mismo nivel de ruido y acoplados a través de un paisaje común, da otra vez un movimiento emparejado, en el espacio de todos los estados juntos.

CO-11 — Averaging agents: exact for identical linear agents, not in general `[PROVED_HERE + STANDARD_FACT]` (EG-08.a)

(a) `PROVED_HERE`. Let every agent have state space R^{n_1} and noise dimension m_1; let A (antisymmetric), G (symmetric, $\succeq 0$), H (symmetric) be constant $n_1 \times n_1$ matrices and Σ a constant $n_1 \times m_1$ matrix with $\Sigma\Sigma^{\mathsf{T}} = 2\varepsilon G$; $J(x) := \tfrac{1}{2}\, x^{\mathsf{T}} H x$, $c \geq 0$, and $J_{\mathrm{int}}(q) := (c/2) \sum_{i<j} |q_i - q_j|^2$. Let $q = (q_1, \ldots, q_N)$ solve the joint paired motion of CO-10 with $A_i = A$, $G_i = G$, $\Sigma_i = \Sigma$ and landscape $\sum_i J(q_i) + J_{\mathrm{int}}(q)$:

```
dq_i = (A − G) ( H q_i + c sum_{j≠i} (q_i − q_j) ) dt + Σ dW_i .
```

Then the average $X := N^{-1} \sum_i q_i$ solves

$$dX = (A - G)\, H\, X\, dt + N^{-1/2}\, \Sigma\, dB, \qquad B := N^{-1/2} \sum_i W_i,$$

where B is a standard Brownian motion. This is the paired motion (M_P) on R^{n_1} with the same ε and data $(A/N, G/N, N \cdot J)$: $(A/N - G/N)\nabla(NJ)(x) = (A - G)Hx$ and $(N^{-1/2}\Sigma)(N^{-1/2}\Sigma)^{\mathsf{T}} = 2\varepsilon(G/N)$. The coupling drops out of the average.

(b) `PROVED_HERE + STANDARD_FACT`. Heterogeneous agents do not average to a motion of this form. For $i = 1, 2$ let x_i solve $dx_i = -\kappa_i\, x_i\, dt + \sigma_i\, dW_i$ (U-15 with $\gamma k = \kappa_i$), with $\kappa_1 \neq \kappa_2$, $\sigma_i > 0$, W_1, W_2 independent, and $x_i(0) \sim N(0, a_i)$, $a_i := \sigma_i^2/(2\kappa_i)$ (the invariant law, U-15 (b)), independent of each other and of W. Then $Y := x_1 + x_2$ is a stationary centered Gaussian process with autocovariance $r(h) = a_1\, e^{-\kappa_1 |h|} + a_2\, e^{-\kappa_2 |h|}$, and for $h > 0$

$$r(2h)\, r(0) - r(h)^2 = a_1\, a_2\, (e^{-\kappa_1 h} - e^{-\kappa_2 h})^2 > 0.$$

A stationary centered Gaussian process with autocovariance r, $r(0) > 0$, that is Markov in its own filtration satisfies $r(t - u)\, r(0) = r(t - s)\, r(s - u)$ for $u \le s \le t$; so Y is not Markov. `STANDARD_FACT`: a solution of a one-dimensional equation $dZ = b(Z)\, dt + \sigma(Z)\, dB'$ with locally Lipschitz b, σ and no explosion is a Markov process [Øksendal 2003, Thm 7.1.2, for globally Lipschitz coefficients; in general, pathwise uniqueness gives uniqueness in law, Karatzas–Shreve 1991, Prop. 5.3.20, and uniqueness in law gives the strong Markov property, ibid., §5.4.C]. Hence neither Y nor the average $Y/2$ has the law of a solution of any one-dimensional equation of the form (M) or (M_P) that satisfies (H). (With $\kappa_1 = \kappa_2 =: \kappa$, Y is again of this form: $dY = -\kappa\, Y\, dt + (\sigma_1^2 + \sigma_2^2)^{1/2}\, dB'$, with the standard Brownian motion $B' := (\sigma_1\, W_1 + \sigma_2\, W_2)/(\sigma_1^2 + \sigma_2^2)^{1/2}$.)

Proof. (a) $\sum_i \sum_{j \neq i} (q_i - q_j) = 0$, since each pair appears once with each sign. Sum the N equations and divide by N. B is a continuous centered Gaussian process with independent increments and $\mathrm{Cov}(B_t) = N^{-1} \cdot N\, t\, I = tI$. The last two identities are direct. (b) For $h \ge 0$, $x_i(t + h) = e^{-\kappa_i h}\, x_i(t) + \int_t^{t+h} e^{-\kappa_i (t+h-r)}\, \sigma_i\, dW_i(r)$, and the integral is independent of $x_i(t)$, so $\mathrm{Cov}(x_i(t + h), x_i(t)) = e^{-\kappa_i h}\, a_i$; the two agents are independent, which gives r. Expanding, $r(2h)r(0) - r(h)^2 = a_1 a_2 (e^{-2\kappa_1 h} + e^{-2\kappa_2 h} - 2e^{-(\kappa_1 + \kappa_2)h})$. Markov identity: for $u < s < t$ the Markov property gives $E[Y_t \mid Y_s, Y_u] = E[Y_t \mid Y_s]$, which for a centered Gaussian vector is $(r(t - s)/r(0))\, Y_s$; multiply by Y_u and take expectations: $r(t - u) = r(t - s)\, r(s - u)/r(0)$. Take $t - s = s - u = h$. ∎

En llano: el promedio de agentes idénticos y lineales vuelve a ser un movimiento de la misma forma (con el paisaje multiplicado por N y la movilidad dividida por N, y el acoplamiento entre ellos desaparece del promedio); el de agentes distintos no: guarda memoria de sus dos ritmos y ya no cabe en una ecuación de un solo estado.

CO-12 — The epistemic-yield principle does not follow from Ax 3.2 and Ax 4.2 `[PROVED_HERE]` (EG-10.c2)

(a) `P_cat(t) = - < grad J(q_base), B(q_base) * u(t) >` (Def 3.1, p.25) is linear in $u(t)$. On an alternative with $u \equiv 0$ (a purely informational action), $P_{\mathrm{cat}} \equiv 0$; so the numerator $E[P_{\mathrm{cat}}]$ of Ax 3.2 (p.27) is 0 for every such alternative, and Ax 3.2 does not rank them.

(b) The decrease of $E_{\mathrm{Hamiltonian}}$ is not determined by the quantities Ax 3.2 and Ax 4.2 depend on: P_{cat}, and S_{total} through L_{physical} and $L_{\mathrm{informational}}$ (D_{met}, the M_{cog}^3 term and the erasure cost $|| dq_{\mathrm{model}}/dt ||^2$, pp.30–31). Example, $d = 2$, on $[0, \pi/2]$: the base state is fixed, with $\Pi(q) = 0$ and $J_\Pi(q)\, p = 0$; $u \equiv 0$; $M_{\mathrm{cog}} = I$; $G_{\mathrm{model}} = I$; $X_H \equiv 0$ on both sides, so $D_{\mathrm{met}} \equiv 0$ (Def 3.2, p.26). Two circles of radius 1, run at unit speed from $(1, 0)$. Path 1, centred at the origin: $q_{\mathrm{est}}(t) = (\cos t, \sin t)$. Path 2, centred at $(2, 0)$: $q_{\mathrm{est}}(t) = (2 - \cos t, \sin t)$. In both, $p_{\mathrm{est}} = \dot{q}_{\mathrm{est}}$ (Ax 6.1 with $M_{\mathrm{cog}} = I$) and $\dot{p}_{\mathrm{est}} = \ddot{q}_{\mathrm{est}}$ have length 1. So the erasure cost is $\equiv 1$ on both paths if q_{model} is q_{est}, and $\equiv 2$ on both if q_{model} is the whole model state $(q_{\mathrm{est}}, p_{\mathrm{est}})$: in either reading the two paths have the same P_{cat} ($\equiv 0$), the same L_{physical} (same base path,

$u \equiv 0$) and the same $L_{\text{informational}}$ ($D_{\text{met}} \equiv 0$, the same M_{cog}, the same erasure cost), so Ax 3.2 and Ax 4.2 give them the same value. But $E_{\text{Hamiltonian}} = \tfrac{1}{2}(|q_{\text{est}}|^2 + |p_{\text{est}}|^2)$ stays 1 along path 1 and goes from 1 to 3 along path 2: the yield principle strictly prefers path 1 (decrease 0 against -2, at equal cost).

(c) Replace, in Ax 3.2, $E[P_{\text{cat}}]$ by the expected decrease of $E_{\text{Hamiltonian}}$ over the horizon, and restrict to alternatives with $u \equiv 0$ whose L_{physical} part of S_{total} is zero. Then Ax 3.2's ratio is the yield of EG-10.c2 (expected decrease per unit of $\beta\, L_{\text{informational}}$). So the principle is Ax 3.2 with a new numerator: an added axiom.

Proof. (a) Def 3.1. (b) $e_q = -q_{\text{est}}$ and $e_p = -p_{\text{est}}$, so $E_{\text{Hamiltonian}} = \tfrac{1}{2}(|q_{\text{est}}|^2 + |p_{\text{est}}|^2)$. Path 1: $|q_{\text{est}}|^2 = 1$ and $p_{\text{est}} = (-\sin t, \cos t)$, so $E_{\text{Hamiltonian}} = 1$. Path 2: $|q_{\text{est}}|^2 = (2 - \cos t)^2 + \sin^2 t = 5 - 4\cos t$ and $p_{\text{est}} = (\sin t, \cos t)$, so $E_{\text{Hamiltonian}} = 3 - 2\cos t$, from 1 at $t = 0$ to 3 at $t = \pi/2$. On both paths $|\dot{p}_{\text{est}}| = |\ddot{q}_{\text{est}}| = 1$. The listed terms of P_{cat} and S_{total} are equal by construction. (c) Substitution. ∎

En llano: que la mejor acción informativa sea la que más baja el error por unidad de coste de pensar no sale de los axiomas 3.2 y 4.2: el 3.2 mide un beneficio que vale cero para cualquier acción puramente informativa y ninguno de los dos mira ese error; hace falta un axioma nuevo.

3. Block C, objects and axioms — the ten answers

3.0 Summary

ID	Question (one line)	Verdict	Bearing statements
EG-02.1	Which printed form of u_{opt} is the minimizer of `Action_Local`?	`FAILS_WITH_COUNTEREXAMPLE`	CO-01; B-08
EG-02.6	`Ricci_Curvature(Error_Tensor)` (§6.5) versus $-2\,R_{ij}$ (App E): which flow; well defined?	`ILL_POSED_AS_WRITTEN`	CO-02, CO-03, CO-04, CO-05
EG-10.a3	Does App E's Step 3 follow from Step 2?	`FAILS_WITH_COUNTEREXAMPLE`	CO-05, CO-03
EG-04.2	What is Frequency(M_{model}) (Def 10.4)?	`ILL_POSED_AS_WRITTEN`	CO-07; B-01, B-07, B-08; U-11
EG-05.b	Is the Beta-Switch (Ax 7.2) an axiom or a corollary?	`HOLDS_UNDER_STATED_HYPOTHESES`	CO-08; CO-01
EG-07.b-src	Regeneration "Regime B only", exemplified by biology	`ILL_POSED_AS_WRITTEN`	CO-09
EG-08.a	Same equations for agent and collective: a self-similarity statement?	`ILL_POSED_AS_WRITTEN`	CO-10, CO-11; U-08, U-11, U-15
EG-10.c2	Is an epistemic-yield principle derivable from Ax 3.2 and Ax 4.2?	`FAILS_WITH_COUNTEREXAMPLE`	CO-12
EG-10.a2 (HOLD)	Is the App E flow a derivable, well-defined object?	`HOLDS_UNDER_STATED_HYPOTHESES` (the flow half, derivable only by the corrected derivation; the composition half has no object and stays with EG-10.a1)	CO-02, CO-03, CO-04, CO-05, CO-06
EG-09.5 (HOLD)	Is the 'work-only' ceiling now settled?	`FAILS_WITH_COUNTEREXAMPLE`	B-10, B-11

Unit IDs of the 2026-08-30 owner-contract intake, as cross-referenced by the triage reviewers: EG-02.1 = EG-02.a, EG-02.6 = EG-02.f, EG-04.2 = EG-04.b.

3.1 EG-02.1 — the sign and form of u_opt

Question (triage `d0cf83e`, exact next question): "With Action_Local(u) = 0.5*||F_req - B u||^2_G_reg + 0.5*lambda_effort*||u||^2 (p.43) and Z = Hessian_u(Action_Local) (Def 7.2, p.44), which of the three expressions equals argmin_u Action_Local, at which evaluation point of the gradient, and is -grad_u S_total (p.8) the same vector as the Thm 7.1 expression?"

GTPA v1.2 locators:

- p.8, summary 6.2: "u_opt = Z^(-1) * (- grad_u S_total)".
- p.43, §7.2: Def 7.1, "F_req = - grad_p(E_Hamiltonian + V_constraint)"; "Action_Local(u) = 0.5 * || F_req - B * u ||^2_G_reg + 0.5 * lambda_effort * || u ||^2".
- p.44: Def 7.2, "Z := Hessian_u(Action_Local) Z = B^T * G_reg * B + lambda_effort * I"; Thm 7.1, "u_opt = Z^(-1) * (Gradient_u(Action_Local))" (no minus sign, also on the rendered page); its Algorithm, step 2 "Invert singular values sigma_i only if sigma_i > epsilon", step 3 "u_opt = V * S_inv
  - U^T * (B^T * G_reg * F_req)".
- Uses without formula: p.45 (Ax 7.2, "|| u_opt ||^2_Metric"); p.73 (App C.4, "Solve u_opt"). Ledger: no claim on u_{opt} (triage).

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`** (Thm 7.1's displayed formula; the other two forms give the minimizer when read as stated below).

Argument.

1. The minimizer. With $G_{\text{reg}} \succeq 0$ and $\lambda_{\text{effort}} > 0$, `Action_Local` is strictly convex with Hessian $Z \succeq \lambda_{\text{effort}} I$ (B-08 (d)) and exactly one minimizer $u^* = Z^{-1} B^{\mathsf{T}} G_{\text{reg}} F_{\text{req}}$ (CO-01 (a)).
2. Step 3's form is u^* whenever the spectral filter removes nothing; every singular value of Z is at least λ_{effort}, so a threshold below λ_{effort} removes nothing (CO-01 (d)). What a larger threshold does, and whether the filtered form is still the minimizer, is EG-03.8 (theorems half), not answered here.
3. Thm 7.1's form, with the gradient taken at a point u_0, is $Z^{-1}\,\nabla AL(u_0) = u_0 - u^*$ (CO-01 (c)). The natural point is $u_0 = 0$, the point from which one Newton step reaches u^*, and there it gives $-u^*$: the opposite vector. It gives u^* only at $u_0 = 2u^*$, a point that presupposes the answer, and at no point fixed without knowing F_{req} (CO-01 (c): $k = N = 1$, $B = G_{\text{reg}} = \lambda_{\text{effort}} = 1$, $F_{\text{req}} = 1$, $u_0 = 0$: the formula gives $-½$, with `Action_Local(-½) = 5/4` against the minimum 1/4 at ½). So of the triage's rival readings, "the gradient meant at a specific point" is true only at $2u^*$, and "a sign typo in Thm 7.1" is the one the mathematics supports.
4. p.8's form uses S_{total}, which Ax 4.2 defines as an integral over the planning horizon (a function of the control path), while Z is the Hessian of `Action_Local`; v1.2 does not say how the gradient of S_{total} becomes a vector in R^k. Read with `Action_Local` in place of S_{total} and the gradient at $u = 0$, it is exactly u^* (CO-01 (b)). In that reading $-\operatorname{grad}_u S_{\text{total}}$ and Thm 7.1's expression are opposite vectors, equal only when $u^* = 0$, that is, when $B^{\mathsf{T}} G_{\text{reg}} F_{\text{req}} = 0$.

Repair adopted: GTPA v2 states one control law, `u_opt := argmin_u Action_Local = Z⁻¹ Bᵀ G_reg F_req = Z⁻¹(−grad_u Action_Local(0))`, with $G_{\text{reg}} \succeq 0$ and $\lambda_{\text{effort}} > 0$ declared. Step 3's form is kept as its computation (no filtering is needed when the threshold is below λ_{effort}; the filtered case is EG-03.8). p.8's summary is rewritten with `Action_Local` in place of S_{total} and the gradient at 0; Thm 7.1's display gets its minus sign.

En llano: de las tres fórmulas de la mejor acción, la del Teorema 7.1 tiene el signo al revés (da justo la acción opuesta a la mejor); la del paso 3 y la del resumen dan la buena si se leen como se dice aquí; la

nueva edición deja una sola.

3.2 EG-02.6 — Ricci in §6.5 versus App E

Question: "(i) Is Ricci_Curvature(Error_Tensor) defined, i.e. is Error_Tensor a metric whose Ricci tensor is meant, and is it G_model? (ii) Does Step 3 (p.78) follow from Step 2 in sign, coefficient and the dropped (1/2tau) G term? (iii) Is the rate -2 or -(1/beta), and on t or t_slow?" (Sub-question (ii) is also the separate unit EG-10.a3, answered in §3.3.)

GTPA v1.2 locators:

- p.7, summary 5.4: "d(G_model)/dt = - 2 * Ricci_Curvature(Error_Tensor)".
- p.41, §6.5, Ax 6.2: "the deformation of the internal metric G_model driven by the curvature of the error stress"; "d(G_model)/dt_slow = - 2 * Ricci_Curvature(Error_Tensor)"; "a slow manifold t_slow, distinct from the fast inference time t"; "this flow is the unique path that minimizes the Fisher-Shannon Action".
- p.77, App E: the "Metric Plasticity" rule "postulated in Chapter 6.5 (dG/dt = -2 Ric)"; Axiom E.1 (the functional W).
- p.78, E.2: Steps 1–3 (§3.3).
- p.79, E.3: "Why does GTPA Chapter 6.5 state simply dG/dt = -2 Ric?"; Thm E.1, "d(G_ij)/dt = -2 * R_ij"; E.4, "dG/dt ~ - (1 / beta) * R_ij".
- "Error_Tensor" occurs only on pp.7 and 41 and is defined nowhere (text sweep). Ledger: no claim on the Ricci flow (triage).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. (i) A Ricci tensor is defined for a metric: a smooth, nondegenerate, symmetric 2-tensor field. "Error_Tensor" is never defined, and Ax 6.2 describes it as "the error stress", not as a metric, so `Ricci_Curvature(Error_Tensor)` has no meaning in v1.2. Read as a fixed metric E other than G_{model}, Ax 6.2 is the flow $\partial_s g = -2\ \text{Ric}(E)$, whose solution is affine in the clock; read as G_{model} itself (or a constant multiple of it), it is Hamilton's Ricci flow of G_{model}. These are different flows: on the sphere they give different metrics at every positive time (CO-04). App E reads §6.5 the second way (p.77, p.79); Ax 6.2's wording does not say which.
2. (ii) Step 3 does not follow from Step 2 as printed: Step 2 has the wrong relative sign between R_{ij} and $\text{Hessian}_{ij}(f)$, Step 3's equation is the one the correct first variation gives but as an ascent, and the dropped metric term only rescales (§3.3; CO-03 (b), CO-05).
3. (iii) With β a positive constant, -2 and $-1/\beta$ are the same flow on two clocks (CO-03 (a)): the coefficient is a choice of time unit. v1.2 writes the flow on t (pp.7, 78–79) and on t_{slow} (p.41) and relates the two clocks only by "distinct", so the rate is not determined by the text; the clock is what must be declared.
4. Well defined? The Ricci flow of a smooth metric is well posed for a short time on a closed manifold, and on R^d for complete metrics of bounded curvature (CO-02 (c)). v1.2's own setting ($M_{\text{model}} \subset T^*Q$, with Q a Sobolev space) is not covered, and v1.2 states no regularity or completeness of G_{model}.
5. The Hessian(f) gap between p.78 and p.79: Thm E.1's reading is right — $\text{Hess}\, f = \tfrac{1}{2} L_{\nabla f}\, g$, and up to the diffeomorphisms generated by ∇f the flow of Step 3 is the Ricci flow (CO-05 (d)). That the agent "absorbs" these diffeomorphisms through Π is v1.2's interpretation, not a mathematical statement.

Repair adopted: if GTPA v2 keeps metric plasticity, it has one metric flow: the Ricci flow $\partial_s g = -2\,\mathrm{Ric}(g)$ of the model metric $g = G_{\mathrm{model}}$, on a declared slow clock s (any constant coefficient is absorbed into the clock, CO-03 (a)), on a closed model manifold or on R^d with a complete metric of bounded curvature (CO-02 (c)). "Error_Tensor" is removed. The phrases "the unique path that minimizes" and "Thermodynamic Necessity" (p.41) and "physically necessary" (p.80) are withdrawn: the flow is a declared modeling choice, with the facts of CO-02, CO-05 and CO-06.

En llano: GTPA escribe de dos maneras que no son la misma la regla con la que cambia la forma del modelo: una usa un "tensor de error" que nunca define y la otra es una regla matemática conocida (el flujo de Ricci) aplicada a la métrica del modelo; la nueva edición, si conserva la regla, se queda con la segunda, con su reloj declarado.

3.3 EG-10.a3 — does App E's Step 3 follow from Step 2?

Question: "Does Step 3 of App E (p.78) follow from Step 2?" (New unit from the triage reviewer's split of EG-10.a2.)

GTPA v1.2 locators:

- p.78, E.2: "the metric G_ij(t) must evolve in the direction of steepest descent of the Action W"; "with respect to an inverse metric perturbation v_ij = delta G_ij"; Step 1 (two variation identities); Step 2, "Grad_G(W)_ij = -R_ij + Hessian_ij(f) - (1/(2*tau)) * G_ij"; Step 3, "we set dG/dt proportional to the negative gradient: d(G_ij)/dt = -2 * (R_ij + Hessian_ij(f))" and "(Ignoring the trace rescaling term for fixed volume)". Signs checked on the rendered page.
- p.77, Axiom E.1: "W(G, f, tau) = Integral_M [tau * (R + ||grad f||^2) + f - d] * u^2 * dV_G", with "u = exp(-f/2)"; p.78: "Minimizing W corresponds to maximizing the entropy of the distribution ...".

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. Sign. Step 2's tensor $-\mathrm{Ric} + \mathrm{Hess}\, f - g/(2\tau)$ has R_{ij} and $\mathrm{Hessian}_{ij}(f)$ with opposite signs; Step 3's $-2(\mathrm{Ric} + \mathrm{Hess}\, f)$ has them with the same sign. No real constant turns one into the other, with or without the metric term: on the unit sphere with $f = z^2$, at a point of the equator, Step 3's tensor is $\mathrm{diag}(-2, -6)$ and Step 2's is $\mathrm{diag}(-1 - a, 1 - a)$, and a real multiple would need $a = -2$, while the printed coefficient is $a = 1/(2\tau) > 0$ and the dropped one $a = 0$ (CO-05 (b)). Since the comparison allows any real factor, it does not depend on whether v is a perturbation of the metric or of its inverse (p.78 says both).
2. Step 2 is not the first variation of W. With the measure $e^{-f}\,dV$ held fixed (v1.2's u^2 as a density), the correct gradient for perturbations of the metric is $-\tau(\mathrm{Ric} + \mathrm{Hess}\, f - g/(2\tau))$ (CO-05 (a); Perelman 2002, §3.1); for perturbations of the inverse metric it changes sign. v1.2 does not fix the variation of f; for any other choice the extra terms are a function times g (CO-05 (a)). So in every reading R_{ij} and $\nabla_i \nabla_j f$ carry the same sign; Step 2 gives them opposite signs, and it lacks the factor τ.
3. Step 3's equation is the right one for the correct variation, up to the metric term, but in the other direction: it is twice the gradient of Perelman's F (and, up to the metric term, $2/\tau$ times the gradient of W) for perturbations of the metric, and along it F does not decrease (CO-05 (c)). "steepest descent" and "Minimizing W" (p.78) have the direction reversed.
4. The dropped term. A term proportional to the metric changes a solution only by a time-dependent constant factor and a change of clock (CO-03 (b)); so dropping it is legitimate up to that rescaling. The reason printed, "for fixed volume", is not the one that makes it legitimate.

5. Coefficient. “proportional” leaves the coefficient free; -2 is a choice of clock (CO-03 (a)).

Repair adopted: GTPA v2 replaces Step 2 by the correct first variation (CO-05 (a)); it states that Step 3’s flow increases F (CO-05 (c)) and equals the Ricci flow up to diffeomorphisms (CO-05 (d)); it gives the rescaling reason for dropping the metric term (CO-03 (b)); and it removes “steepest descent”.

En llano: en el apéndice E el paso 3 no se deduce del paso 2, porque el paso 2 trae cambiado el signo entre la curvatura de Ricci y el término de f; la ecuación del paso 3 sí es la conocida, pero hace subir, no bajar, la cantidad F de Perelman (y la W que el texto dice minimizar sube cuando además τ decrece, Perelman §3.1).

3.4 EG-04.2 — Frequency(M_model)

Question: “What map sends a manifold with metric and potential (M_model, G_model, M_cog) to a frequency spectrum? Under what spectral distance is ’Frequency(M_model) = Frequency(M_base)’ meaningful when dim M_model != dim M_base?”

GTPA v1.2 locators:

- p.58, Def 10.4: “Resonance Condition: Frequency(M_model) = Frequency(M_base).”; the “Effect” clause (“Power_Factor -> 1”). “Frequency(“ occurs only here (triage).
- p.46, Def 7.3: “a characteristic frequency spectrum omega_base derived from the eigenvalues of the Hessian H(q)”; §7.5: “min_omega || omega_ctrl - omega_base ||” (`omega_ctrl` occurs only here and is undefined).
- p.21, Ax 2.1: “a Symplectic Form omega_model homomorphic to the external form omega_ base”: the same name `omega_base` for another object.
- p.37, Ax 6.1 (model dynamics); p.34, Def 5.1 ($E_{\text{Hamiltonian}}$); Block B, B-08 (EG-04.1).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. A manifold alone has no frequencies. For a linear second-order system the mode frequencies are Spec(M, H), the square roots of the eigenvalues of $M^{-1}H$, where M multiplies the second derivative and H is the stiffness at an equilibrium (CO-07 (a); B-08 (b)). Spec does not depend on the coordinates but depends on both matrices; with one fixed it takes every value. So “Frequency(M_model)” needs a declared pair (M, H) and a declared equilibrium; Def 7.3’s “eigenvalues of the Hessian H(q)” names only H, as Block B noted for EG-04.1.
2. With the triage’s data $(M_{\text{model}}, G_{\text{model}}, M_{\text{cog}})$, v1.2 still does not fix the map. M_{cog} supplies M (Block B repair). The stiffness comes from Ax 6.1’s term `F_cognitive = - grad( beta * E_ Hamiltonian )`, and v1.2 does not say which gradient: the differential in q_{est} gives $H = \beta\, G_{\text{model}}$, the gradient for the metric G_{model} gives $H = \beta\, I$, and the two spectra differ (CO-07 (b)). The first is the reading in which $F_{\text{cognitive}}$ has the type of $p_{\text{est}} = M_{\text{cog}}\, \dot{q}_{\text{est}}$.
3. On the base side the UFD motion is first order (C5): at a nondegenerate minimum its linearization has only real eigenvalues unless there is a circulating part (CO-07 (c); B-08 (c)). So the gradient base motion has no frequencies to match; a second-order base system is a separate object (U-11).
4. With $\dim M_{\text{model}} \neq \dim M_{\text{base}}$ the two multisets have different sizes and are never equal; equality as sets, containment, and equality of the lowest or of the highest frequency give different answers on the same pair (CO-07 (d)). v1.2 names no distance, so the condition has no meaning until one relation is declared.

5. The name `omega_base` denotes a symplectic form on p.21 and a frequency spectrum on p.46.

Repair adopted: GTPA v2 does not use Frequency(·) of a manifold. For a declared linear or linearized second-order system, with declared $M \succ 0$ and declared stiffness $H \succ 0$ at a declared equilibrium, it defines the mode spectrum Spec(M, H) (CO-07 (a)); on the model side of Ax 6.1, with constant coefficients and the differential reading of "grad", this is Spec$(M_{\text{cog}}, \beta G_{\text{model}})$. For the first-order base it uses the eigenvalues of the linearization (rotation rates exist only with a circulating part). Def 10.4's condition is dropped or replaced by one declared relation between the two multisets (for instance containment), with no "Effect" claimed beyond B-08. `omega_base` is renamed on one of its two pages.

En llano: GTPA habla de "la frecuencia" del modelo y del mundo sin definirla; una forma sola no tiene frecuencias: hacen falta dos matrices en un punto de equilibrio, y GTPA no fija una de ellas; la base de primer orden no oscila salvo que haya giro; y con tamaños distintos "igualar frecuencias" admite varias lecturas con respuestas distintas.

3.5 EG-05.b — is the Beta-Switch an axiom or a corollary?

Question: "Does the switching rule follow from minimizing S_total (Ax 4.2) over a binary mode variable, with Unified_Cost_Inference (p.45) and Unified_Cost_Control = ||u_opt||^2_Metric (Thm 7.1)? If yes, which beta placement results? The hysteresis clause is excluded (EG-05.c)."

GTPA v1.2 locators:

- p.45, Ax 7.2: "Unified_Cost_Inference = beta * || dp_est/dt ||^2"; "Unified_Cost_Control = || u_opt ||^2_Metric"; "IF Cost_Control < Cost_Inference: Mode = ACTION"; the hysteresis clause (excluded: EG-05.c, terminal HOLD_SOURCE_UNDERSPECIFIED; C0137 held).
- p.44, §7.3: "The Agent minimizes the Unified Action S_total. However, computational resources (Information) and physical power (Work) are distinct reservoirs."
- p.8, summary 6.3: "Mode = (Cost_Control < beta * Cost_Inference) ? ACTION : INFERENCE".
- pp.30–31, Ax 4.2: "The Agent evolves to minimize the functional S_total over its planning horizon T_H"; its $L_{\text{informational}}$ lists "Erasure Cost: The cost of updating beliefs (|| dq_model/dt ||^2)".
- The locator is closed by EG-05.a (PASS_LOCATOR_GTPA_V1_2_ONLY). Ledger: no claim on the Beta-Switch (triage).

Verdict: **`HOLDS_UNDER_STATED_HYPOTHESES`**.

Argument.

1. The mode choice as a minimization. Take a mode path $m(t) \in \{\text{act}, \text{inf}\}$ whose running cost is the control cost in `act` and β times the inference cost in inf (the question's costs, with Ax 4.2's weight). If these costs do not depend on earlier choices (hypothesis (X)), the path that picks at each time the smaller of c_{act} and $\beta\, c_{\text{inf}}$ minimizes the integral, and every minimizer agrees with it wherever the two differ (CO-08 (a)). That is the switching core of Ax 7.2: under (X) it is a corollary of minimizing the cost over the mode, not an independent axiom.
2. β placement. Under (X), β enters once, on the inference side (CO-08 (b)). p.45's placement is that rule. p.8's placement is the same rule if its `Cost_Inference` is the cost without β; read with p.45's `Unified_Cost_Inference`, it counts β twice and decides differently ($\beta = 2$, $c_{\text{inf}} = 1$, $c_{\text{act}} = 3$).
3. What (X) excludes. Ax 4.2 minimizes over the planning horizon, and there the state carries the effect of earlier modes: acting changes q, inferring changes q_{est}, and u_{opt} depends on F_{req}, hence on the error (CO-01). Without (X) the instantaneous rule need not minimize (two-step witness,

CO-08 (c)). So Ax 7.2 does not follow from Ax 4.2 in general; outside (X) it is a declared one-step policy.

4. Two typing points that the hypotheses make explicit: $|| \, dp_{\text{est}}/dt \, ||^2$ is not the erasure cost $|| \, dq_{\text{model}}/dt \, ||^2$ that Ax 4.2 lists (with `dq_est/dt = M_cog^(-1) * p_est`, one depends on the rate of p_{est}, the other on p_{est}); and "Metric" in $|| \, u_{\text{opt}} \, ||^2_{\text{Metric}}$ is undeclared. Both costs are taken as given, as the question stipulates.

Repair adopted: GTPA v2 states the switching core as a corollary (CO-08 (a)) under declared hypotheses: exclusive modes; running costs $|| \, u_{\text{opt}} \, ||^2_{\text{Metric}}$ (with the metric declared) and $\beta \, || \, dp_{\text{est}}/dt \, ||^2$; and (X). β is written once, as on p.45. Outside (X) the rule is labelled a one-step policy, not a consequence of Ax 4.2. The hysteresis clause stays with EG-05.c.

En llano: la regla de elegir entre actuar o pensar según qué sale más barato se deduce de minimizar el coste total cuando lo que se elige ahora no cambia los costes de después; entonces el factor β va una sola vez, del lado de pensar; si los costes sí dependen de lo elegido antes, la regla puede fallar.

3.6 EG-07.b-src — "Regime B only" and the biological example

Question: "In GTPA v1.2, is regenerative recovery (Work_Term with eta_regen>0, p.46) derivable only under the logarithmic-barrier potential of Regime B (p.35)? If so, is the biological example inconsistent with pp.56 and 75, or does a Regime A version exist?" (A producer-originated unit; the triage reviewer places it with EG-02, the internal inconsistencies of the work.)

GTPA v1.2 locators:

- p.46, §7.4: "Regenerative Agency (Regime B only): A Robust Agent can act as a generator. During the braking phase, it can recover a fraction eta_regen of the kinetic energy back into its metabolic store (e.g., regenerative braking in robotics, ATP recovery in biology)."; "Work_Term = (<F, v> > 0) ? <F, v> : (1 - eta_regen) * <F, v>" (checked on the rendered page).
- p.34, Def 5.2, "Regime A: The Soft Potential (Class A)"; p.35, "Regime B: The Logarithmic Barrier (Class B)", and the Regime A animal example.
- p.55: "Class B (Robust Artificial)"; p.56: "constraints that are physically absent in biology"; p.75, D.2: "Class A agents model humans and animals".

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. Not "derivable only under Regime B". v1.2 gives no derivation of regeneration. Its formula is a function of `<F, v>` and `eta_regen` alone; no constraint potential enters it, so it is the same with the soft potential of Regime A and with the logarithmic barrier of Regime B (CO-09 (a)). "Regime B only" is a label, not a consequence, and a Regime A version is the same formula.
2. The label contradicts the example. With Regime B = Class B (p.35) and biology outside Class B (pp.56, 75), the biological example makes four sentences of v1.2 jointly inconsistent (CO-09 (c)). Of the triage's two rival readings, "an editorial example with no eligibility claim" removes the contradiction; "«Regime B» meaning something other than Class B at p.46" is contradicted by p.46 itself ("A Robust Agent"; "Robust Artificial" is Class B, p.55).
3. The formula does not say what its sentence says: read as a cost it credits the fraction `1 − eta_regen`, not `eta_regen`, of `|<F, v>|` on phases with `<F, v> < 0`; read as a loss it has the wrong sign there (CO-09 (b)).

Repair adopted: in GTPA v2, `eta_regen ∈ [0, 1]` is a declared parameter of the running cost for any agent, with no regime restriction, and the cost is `<F, v>⁺ − eta_regen · <F, v>⁻` (CO-09 (b)). The biological example is dropped: GTPA v2 makes no empirical claim about any kind of agent.

En llano: el término de "regeneración" no usa para nada el tipo de barrera, así que no hay nada que lo haga "solo del régimen B"; el ejemplo de la biología contradice lo que el propio texto dice de la biología; y la fórmula, leída como coste, recupera la fracción contraria a la que dice el texto, y leída como pérdida tiene el signo cambiado.

3.7 EG-08.a — the agent/collective nesting

Question: "Is there a coarse-graining map Phi from N agents, each satisfying Ax 1.2 and Ax 4.2, to a single tuple (Pi, Theta, H_int) (Def D.1 p.74) whose image again satisfies Ax 1.2 and Ax 4.2, with Cost_Collective (Thm 9.1) as its physical cost term? If no such Phi exists, «autosimilitud estructural» has no formal referent in the work." (Triage reviewer: the label and the question are right; the claim that the premise is inaccurate "goes too far", since Ch 8 reuses the p-equation form and the phase vocabulary returns in Ch 9.)

GTPA v1.2 locators:

- p.17, Ax 1.2 (the first-order master equation); p.30, Ax 4.2.
- pp.48–50, Ch 8: Ax 8.1; Def 8.1; the follower equation "dp_F/dt = - grad(J_self) - grad(V_ influence)" (p.49); Thm 8.1 (p.50).
- pp.51–54, Ch 9: Ax 9.1; Def 9.1 (C_{ij}); Thm 9.1, "Cost_Collective ~ || Sum(v_i) ||^2" (p.52); Thm 9.2–9.3; the phases (pp.53–54).
- p.74, Def D.1: "An Agent A is defined by the tuple (Pi, Theta, H_int)."
- No fractal, self-similarity or renormalization mathematics in v1.2 (triage, text sweep). Ledger: GTPA-C0056 (ADMITTED, DEF, NONPROBATIVE) records agent count and topology as composition regimes; no admitted claim states a self-similarity or cross-scale law.

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. The Ax 4.2 half of the question cannot be posed in v1.2: no collective M_{model}, Π, Θ, H_{int} or S_{total} is defined (H_{int} is named on p.52 and never written), and `Cost_Collective` is given only up to "~". So "an image that satisfies Ax 4.2 with Cost_Collective as its cost term" has no objects to be checked against.
2. The premise of the demand, "same equation families in Ch 1 and Ch 9", does not hold on reading. Ch 9 restates no equation of motion. Ch 8's follower equation is an equation for dp_F/dt (second order in the position), with no noise, a different object from the first-order Ax 1.2 (U-11 (a)). The shared vocabulary (phases indexed by T_{info}) is vocabulary.
3. What is true, for the Ax 1.2 half, on the UFD base:
   - joining: N agents paired with the same ε, coupled through a joint landscape, form one paired motion on the joint state space (CO-10) — the same equation at the level of the joint state, with no reduction;
   - exact reduction in a special case: the average of N identical linear paired agents, even with a pairwise consensus coupling, is again a paired motion with the same ε, the landscape multiplied by N and the mobility and circulating part divided by N (CO-11 (a));

- no reduction in general: the sum of two independent Ornstein–Uhlenbeck agents with different rates is not a Markov process, so it is not a motion of the form (M) or (M_P) in its own variable (CO-11 (b)). So a coarse-graining that sends agents to one agent of the same form exists in a special case and fails in general; "the collective is structurally self-similar to the agent" is false as a general law.

Repair adopted: GTPA v2 states CO-10 and CO-11 as the exact content of "the same equation at two levels" and makes no structural self-similarity statement. A cross-scale law would need a declared coarse-graining with an exact or bounded commuting statement; none is admitted in the lab.

En llano: juntar varios agentes da otra vez la misma ecuación en el espacio de todos juntos, y el promedio de agentes iguales y lineales también; pero el promedio de agentes distintos ya no, así que la "autosimilitud" entre agente y colectivo no es un enunciado de GTPA.

3.8 EG-10.c2 — an epistemic-yield principle

Question: "From Ax 3.2 (p.27) and Ax 4.2 (pp.30-31), can one derive, as a special case or corollary, that the optimal informational action maximizes the expected decrease of E_Hamiltonian (Ch.5, p.33ff) per unit of beta*L_informational over the horizon T_H? Or does the derivation need an extra axiom that identifies P_cat with error reduction?"

GTPA v1.2 locators:

- p.27, Ax 3.2: "eta_est(t) = E[P_cat(t)] / E[S_total(t)]", with "S_total = Work_Physical + beta * D_met".
- p.25, Def 3.1: "P_cat(t) = - < grad J(q_base), B(q_base) * u(t) >".
- pp.30–31, Ax 4.2: "S_total = Integral [L_physical(q, u) + beta * L_informational(q_model)] dt", with $L_{\text{informational}}$ made of D_{met}, the M^3_{cog} penalty and the erasure cost.
- p.26, Def 3.2 (the error of D_{met}, a mismatch of drifts); p.33, Axiom 5.1, and p.34, Def 5.1 ($E_{\text{Hamiltonian}}$, built from another error, (e_q, e_p)); p.40, Thm 6.2 (T_H). S_{total} is written two ways (p.27, p.30).

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. Ax 3.2 cannot rank informational actions: its numerator $E[P_{\text{cat}}]$ is linear in u and is zero on every purely informational action (CO-12 (a)).
2. $E_{\text{Hamiltonian}}$ enters neither axiom: D_{met} is built from the drift mismatch of Def 3.2, not from $E_{\text{Hamiltonian}}$. Two model-side paths with the same P_{cat} and the same terms of S_{total} can have different decreases of $E_{\text{Hamiltonian}}$; the yield principle strictly prefers one of them, while both axioms give them the same value (CO-12 (b): two unit circles, equal costs whether the erasure cost is read on q_{est} or on the whole $(q_{\text{est}}, p_{\text{est}})$). So the principle is neither a special case nor a corollary of Ax 3.2 and Ax 4.2.
3. What it needs: replacing $E[P_{\text{cat}}]$ in Ax 3.2 by the expected decrease of $E_{\text{Hamiltonian}}$, for actions with no L_{physical} part, turns Ax 3.2's ratio into the yield (CO-12 (c)). That replacement is the extra axiom the question names ("identifies P_cat with error reduction"). Whether FORMACION then calls the principle a declared analogy is its own label, not issued here.

Repair adopted: GTPA v2 does not state an epistemic-yield principle as derived. If it keeps one, it is a declared principle — Ax 3.2 with the numerator replaced as in CO-12 (c) — and it needs an object for "informational action" (EG-10.c1 is `UNMAPPED_TO_GTPA`).

En llano: que la mejor acción informativa sea la que más baja el error por unidad de coste de pensar no sale de los axiomas 3.2 y 4.2: el 3.2 mide un beneficio que vale cero para cualquier acción puramente informativa, y ninguno de los dos mira ese error; hace falta un axioma nuevo.

3.9 EG-10.a2 (HOLD, follows EG-02.6) — the App E flow as an object

Question. Demand: "Is the App E smoothing flow itself a derivable, well-defined object that zoomout could compose with (EG-08)?" Exact next question after the triage reviewer's split: "(1) Is the flow the Ricci flow of G_model (App E) or a flow driven by the 'Ricci curvature of the error tensor' (Ax 6.2)?" (Blocker: EG-02.6, answered in §3.2. The reviewer dropped EG-08 as a blocker; question (2) became EG-10.a3.)

GTPA v1.2 locators: p.41 (Ax 6.2); pp.77–80 (App E; Thm E.1, p.79; E.4, pp.79–80).

Verdict: **`HOLDS_UNDER_STATED_HYPOTHESES`** (for the flow App E means, read as below; the composition clause has no object yet).

Argument.

1. Which flow: App E's, the Ricci flow of G_{model} (Thm E.1, p.79), which is the one GTPA v2 adopts (§3.2). Ax 6.2's `Ricci_Curvature( Error_Tensor )` is ill-posed (§3.2) and, read as the Ricci tensor of another metric, is a different flow (CO-04).
2. Well defined: for a smooth G_{model} on a closed model manifold, or a complete one of bounded curvature on R^d, the flow exists and is unique for a short time (CO-02 (c)), on a declared clock (CO-03 (a)). No smoothing property is claimed here beyond short-time existence and uniqueness: the "Geometry Smoothing" of Ax 6.2 (p.41) is not proved, and CO-06 shows that the flow can end in finite time with curvature tending to infinity. On a flat model metric (for instance the one that a diffeomorphism of R^d onto itself carries from the Euclidean metric) $\mathrm{Ric} \equiv 0$, and the flow does nothing.
3. Derivable: yes, but not by App E's printed steps (§3.3). Up to diffeomorphisms it is the flow $-2(\mathrm{Ric} + \mathrm{Hess}\, f)$, which is twice the gradient of Perelman's F and increases it (CO-05 (c), (d)).
4. "that zoomout could compose with": neither v1.2 nor GTPA v2 has a formal object for zoomout (EG-10.a1 is `UNMAPPED_TO_GTPA`: a coarse-graining of representations or a time step of dG_{model}/dt, not decided). The Ricci flow is a one-parameter family of metrics on one manifold, not a map between representations, and EG-08.a found no self-similarity statement (§3.7). There is nothing yet to compose; nothing here says that zoomout is a step of this flow.

Repair adopted: as in §3.2 and §3.3; the composition question stays with EG-10.a1.

En llano: el flujo del apéndice E, leído como el flujo de Ricci de la métrica del modelo, está bien definido durante un tiempo en los casos estándar y se puede deducir (aunque no como lo hace el apéndice), pero no siempre suaviza, porque puede acabar en tiempo finito; y todavía no existe ningún objeto "zoomout" con el que componerlo.

3.10 EG-09.5 (HOLD, follows EG-09.3) — the 'work-only' ceiling

Question. Demand: "If the tranche is not viable, declare the 'work-only' ceiling permanent." Exact next question: "After an EG-09.3 evaluator result exists (or Chris explicitly declines to reopen), does FORMACION record the ceiling as 'work-only / no admitted stochastic claim' with the date and the reason?" (Triage reviewer's corrected closure: "no admitted ledger claim covers SDE, Langevin or FDT dynamics on a continuous state space".)

Locators: no v1.2 object. Block B §3.7 (EG-09.3), B-10, B-11; ledger GTPA-C0115 (ADMITTED; its assumptions exclude the stochastic lane) and C0118, C0129, C0132 (ADMITTED; finite-state or sampling objects).

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`** (the ceiling as worded).

Argument.

1. The trigger has occurred: Block B answered EG-09.3 `HOLDS_UNDER_STATED_HYPOTHESES` (without a compact tube), and an independent same-provider read of it exists (Block B status: FIX_NEEDED, both findings fixed, not re-read).
2. The condition of the demand is false: the stochastic efficacy question has exact statements (B-10: representation, first variation, short-time limit, invariance under changes of coordinates; B-11: a lift through the paired mobility). So no permanent 'work-only' ceiling is declared.
3. The wording is inaccurate on both counts: 'work-only' — formal statements now exist in the GTPA v2 text (B-10, B-11); 'no admitted stochastic claim' — GTPA-C0132 (ADMITTED) concerns the stochastic evolutions of finite generators. The accurate statement is the reviewer's.

Settled, and how. What the crosswalk gives FORMACION to record, dated 2026-09-26, reason "EG-09.3 answered in GTPA v2 Block B": the stochastic efficacy of an action on the base motion has formal statements in the GTPA v2 text (B-10, B-11), at the ceiling FORMAL_LOCAL_CONDITIONAL / NONPROBATIVE, with one same-provider read; none of them is a ledger claim; no admitted ledger claim covers SDE, Langevin or FDT dynamics on a continuous state space; the ceiling moves only when B-10 or B-11 passes the ledger gate (from C0138).

En llano: el techo "solo la obra" ya no vale, porque la pregunta del bloque B tiene respuesta bajo sus hipótesis; lo que se anota, con fecha 26-09-2026, es que ya hay enunciados formales en el texto nuevo, pero ninguno admitido todavía en el registro de claims.

4. What this file does not say

- No physics: no temperature, mass, energy, power, force, momentum, impedance, resonance, entropy or metabolism as an object; v1.2's names are used as names, and Perelman's functionals are functionals.
- No GTPA claim: nothing enters the ledger and no claim ID is assigned.
- No MATH validity: no MATH object is used and no binding is created. No TECD text was read.
- No UFD statement is added or changed; M_{cog}, Spec, the model-side metric flow and the mode switch are GTPA-side objects.
- Nothing about persons, animals, biology, minds or agents in the world; no empirical claim.
- Not answered here: the theorems half of Block C (EG-03.1–03.11 and the Thm 9.1 half of EG-03.0), including EG-03.8 (the filtered pseudo-inverse); EG-10.a1 and EG-10.c1 (`UNMAPPED_TO_GTPA`); EG-05.c (hysteresis, terminal HOLD); the Ricci flow on v1.2's Sobolev setting, or outside the classes of CO-02 (c).

5. Open points (not asserted)

- Whether GTPA v2 keeps a metric flow on the model side at all, and on which class of model manifolds (closed, or R^d with complete metrics of bounded curvature).

- The typing of "grad" in $F_{\text{cognitive}}$ (differential or metric gradient), which fixes the model-side stiffness (CO-07 (b)).
- Which relation, if any, replaces Def 10.4's "=" (CO-07 (d)).
- A cross-scale law beyond CO-10 and CO-11 (none is proposed).
- A re-read of this file after the fixes of its one independent read (not done).

6. Sources and references

Pins: research `7db1e5ef1e1e04daafab7b62538068afafd5dbf5`; UFD nucleus `ufd/UFD-NUCLEO.md` (blob `4f52a62c`); Block B `gtpa-v2/BLOCK-B-ANSWERS.md` (blob `3d8dea16`); crosswalk `gtpa-v2/CROSSWALK-v1.2.md` (blob `3af9f8d9`); triage `d0cf83e3f1e7ac50115509f32d00aa58ed0bd51a` (`handoffs/governance/2026-09-25-formacion-eg01-eg10-eg19-triage-01.md`, blob `00a1e1eb`; .json, blob `a772d5cf`); architecture proposal `handoffs/governance/2026-09-25-theory-architecture-proposal-v1.md` (blob `7fc2974c`); ledger `gtpa-lab/CLAIM-LEDGER.md` (blob `84b445d6`); GTPA v1.2 text `quasar:modulos/formacion/fuentes/GTPA-v1.2.txt` (blob 32787369) and PDF (SHA-256 `0c9434c1c6f687e1e16100929641a1a81de0a0c9ac808d18b6991eb9cf75865f`).

Textbooks and papers (locators as used above):

- B.-L. Chen, X.-P. Zhu, "Uniqueness of the Ricci flow on complete noncompact manifolds", *J. Differential Geom.* 74 (2006) 119–154.
- D. M. DeTurck, "Deforming metrics in the direction of their Ricci tensors", *J. Differential Geom.* 18 (1983) 157–162.
- M. P. do Carmo, *Riemannian Geometry*, Birkhäuser, 1992 — Ch. 2, Ch. 4, Ch. 8.
- R. S. Hamilton, "Three-manifolds with positive Ricci curvature", *J. Differential Geom.* 17 (1982) 255–306.
- I. Karatzas, S. E. Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed., Springer, 1991 — Prop. 5.3.20; §5.4.C.
- B. Øksendal, *Stochastic Differential Equations*, 6th ed., Springer, 2003 — Thm 7.1.2.
- G. Perelman, "The entropy formula for the Ricci flow and its geometric applications", arXiv:math/0211159 (2002) — §1, §1.1, §3.1.
- W.-X. Shi, "Deforming the metric on complete Riemannian manifolds", *J. Differential Geom.* 30 (1989) 223–301.

- The UFD nucleus's references (UFD §9) and Block B's (Block B §6) for the statements U-xx and B-xx cited here.